HOMEWORK 9.4 · UNIT 9 · NC.8.G.9

# Unit 9 Review

May 6 / May 7 · Targets: G.9-2, G.9-3, G.9-5, G.9-4, G.9-6, G.9-8, G.9-7, G.9-9, G.9-10

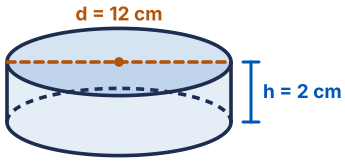
Name _____ Date _____ Block _____

**Remember:** Cylinder $V = \pi r^2 h$ · Cone $V = \frac{1}{3}\pi r^2 h$ · Sphere $V = \frac{4}{3}\pi r^3$. Hemisphere = $\frac{1}{2}$ of a sphere. $r = d \div 2$. Answer in terms of π, or use 3.14 and round. Always use cubic units. A cone = $\frac{1}{3}$ of its cylinder. A sphere = $\frac{2}{3}$ of the cylinder around it.

**SCORE 3.0** Approaching · the basics

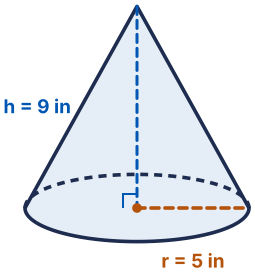
Targets: G.9-2, G.9-3, G.9-5, G.9-4, G.9-6

**1** Find the volume. Answer in terms of π. REVIEW 9.2 G.9-2

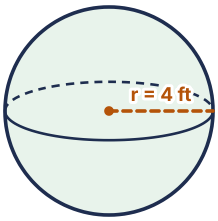



(A) $72\pi$ cm$^3$
(B) $288\pi$ cm$^3$
(C) $24\pi$ cm$^3$
(D) $12\pi$ cm$^3$

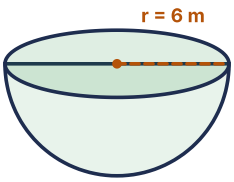
**2** Find the volume of the cone. Use 3.14 for π. Round to the nearest tenth. REVIEW 9.1 G.9-3,G.9-5




**3** Find the volume of the sphere. Answer in terms of π. REVIEW 9.1 G.9-4




**4** Find the volume of the hemisphere. Use 3.14 for π. Round to the nearest tenth. REVIEW 9.3 G.9-4,G.9-5




**5** A cone holds one-third as much as a cylinder with the same base and height. REVIEW 9.1 G.9-6

True False

**SCORE 4.0** Proficient · on target

Targets: G.9-8, G.9-7

**6** A cylinder holds $245\pi$ cm$^3$. Its height is 5 cm. Find its radius.

REVIEW 9.2 G.9-8

**7** **Dev** made a mistake. Circle it. Explain it. Fix it.

REVIEW 9.2 G.9-7

DEV'S WORK

Sphere: $r = 3$ in

$V = \frac{4}{3}\pi(3)^2$

$V = 12\pi$ in$^3$

**8** A traffic cone has a diameter of 10 in and a height of 18 in. Find its volume. Use 3.14 for π. Round to the nearest tenth.

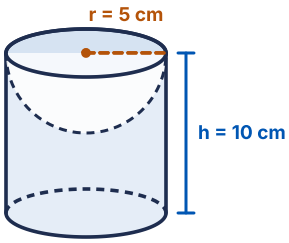
REVIEW 9.2 G.9-7

**SCORE 5.0** Mastery · go deeper

Targets: G.9-9, G.9-10

**9** A block is a cylinder ($r$ = 5 cm, $h$ = 10 cm). A hemisphere with $r$ = 5 cm is scooped out of the top. How much is left? Answer in terms of π. Then round.

REVIEW 9.3 G.9-9




Cylinder = _____ . Hemisphere = _____ . Left = _____ .

**10** Cone A has $r$ = 2 and $h$ = 3. Cone B triples the height. Cone C triples the radius. How many times more does each hold? Explain.

REVIEW 9.3 G.9-10

B holds _____ times. C holds _____ times, because _____ .